# TCT-382 Data Science Applications
# QUIZ

Name: ID:

Q1. Data science is best described as:

(A) A field focused only on storing large databases

(B) An interdisciplinary field that uses statistics, computing and domain knowledge to extract insights from data

(C) The study of designing computer hardware

(D) The practice of creating charts only

Q2. The first step in a typical data science problem-solving workflow is:

(A) Deploying the model

(B) Visualizing the final results

(C) Defining the problem and understanding the domain

(D) Writing the final report

Q3. Which pandas function is used to read a CSV file into a DataFrame?

(A) pd.open_csv()

(B) pd.load_text()

(C) pd.read_csv()

(D) plt.read()

Q4. Which of the following is an example of a primary data source?

(A) A government report published last year

(B) A survey conducted by the researcher themselves

(C) A dataset downloaded from Kaggle

(D) A table copied from a textbook

Q5. JSON is best described as:

(A) A spreadsheet-only format

(B) A binary image format

(C) A lightweight text-based key-value format commonly returned by web APIs

(D) A data compression algorithm

Q6. Data munging (wrangling) refers to:

(A) Transforming raw data from one form into another, more usable format

(B) Encrypting data for transmission

(C) Deleting an entire dataset

(D) Backing up data to the cloud

Q7. For a numeric column containing extreme outliers, which imputation value for missing entries is generally most robust?

(A) Mean

(B) Median

(C) Maximum

(D) Minimum

Q8. In pandas, df.dropna() will:

(A) Fill all missing values with zero

(B) Remove all columns

(C) Remove rows containing missing values

(D) Sort the DataFrame

Q9. Which expression filters rows of a DataFrame df where the column 'age' is greater than 30?

(A) df[df['age'] > 30]

(B) df.filter('age' > 30)

(C) df.drop(age > 30)

(D) df['age'] = 30

Q10. Using min-max normalisation, a value of 30 from a feature with minimum 10 and maximum 50 is scaled to:

(A) 0.25

(B) 0.75

(C) 1.0

(D) 0.5